Soil Advances 6 (2026) 100131

Contents lists available at ScienceDirect

Soil Advances

journal homepage: www.sciencedirect.com/journal/soil-advances

Operationalization of digital soil mapping in detailed soil survey-establishing the concept and practical feasibility in the Eastern Ghat Highlands of India

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ARTICLE INFO

Keywords:
Random Forest Classification model
Terrain Mapping Units
Soil Management Units
Bayesian Structural Equation Model
Eastern Ghat

ABSTRACT

The paradigm shift from *conventional* soil survey to Digital Soil Mapping (DSM) holds immense potential, yet its operationalization at the soil series level remains underexplored. This study pioneers the application of high-resolution (5 m) DSM for categorical soil series delineation in the geomorphologically complex Eastern Ghat Highlands, India- highlighting the potential of machine-learning-based approaches to support detailed soil survey in complex terrains. A dual-framework was employed, integrating *Terrain Mapping Unit (TMU)*-based *conventional* surveys with Random Forest (RF) classification model and 35 environmental covariates derived from remote sensing, DEMs, and climatic datasets. The model was calibrated on 60 soil profiles optimally selected using the *Conditioned Latin Hypercube Sampling (cLHS)* algorithm. Predictive performance was evaluated against 185 field-observed soil profiles withheld from RF calibration, while pixel-to-pixel comparison with the conventional soil map was treated separately as a spatial-concordance assessment. Model uncertainty was quantified using Shannon entropy. *Conventional* mapping identified 22 soil series over 20,722 ha, while the *RF* model achieved outstanding predictive accuracy (97.7%) and strong spatial concordance with $< 20\%$ areal deviation for most series. Variables importance analysis highlighted TMU, landform, and land use/land cover as dominant predictors, and Bayesian Structural Equation Modelling confirmed relief ($B = 0.80$) as the key pedogenic driver. Both approaches yielded 13 *Soil Management Units (SMUs)* with high agreement ($K = 0.811$), twelve of which exhibited near-perfect spatial overlap. Notably, DSM enhanced landscape discrimination in inaccessible forested terrain, revealing pedological heterogeneity that is often masked in polygon-based mapping. By bridging predictive modelling with process-pathway analysis, the study establishes a replicable and scalable framework for series-level soil mapping. The findings demonstrate that DSM not only replicates but also enhances conventional surveys, offering actionable, raster-based spatial frameworks for precision agriculture, soil conservation, and climate-resilient land-use planning in data-scarce regions worldwide.

1. Introduction

Soil is a fundamental natural resource, forming the basis of terrestrial ecosystems and sustaining human life through its diverse functions (Dominati et al., 2014; Powlson et al., 2016). Soil variability, driven by factors like climate, topography, parent material, and biological activity, necessitates its systematic classification to understand its properties and suitability for various applications (McBratney et al., 2000; Brevik et al., 2020). Soil classification is the process of organizing soils into categories based on their physical, chemical, and biological properties (Krasilnikov et al., 2009). Globally, two prominent systems are widely used: the USDA Soil Taxonomy and the World Reference Base for Soil Resources (WRB) (Nachtergaele et al., 2000). The USDA Soil Taxonomy employs a hierarchical framework, classifying soils from broad categories of orders and suborders to soil series at the tip of the pyramid, based on properties such as texture, mineralogy, and moisture regime (Rossiter, 2007; Kyebogola et al., 2020). The WRB emphasizes key diagnostic horizons and soil properties. While both international and local systems aid

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https://doi.org/10.1016/j.soilad.2026.100131
Received 19 January 2026; Received in revised form 16 July 2026; Accepted 16 July 2026
Available online 17 July 2026
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research and management, regional adaptations address unique soil traits for practical applications such as crop suitability and erosion risk. However, mismatches between local and global systems highlight the need for harmonized, standardized approaches (Nachtergaele et al., 2010; Zádorová et al., 2021).

The determination of soil series has traditionally relied on extensive field surveys, laboratory analyses, and expert opinion-based interpretations (McBratney et al., 2000; Tan, 2005). This process, while robust, is resource-intensive, requiring significant time, labour, and financial investment. Moreover, *conventional* methods often produce polygon-based maps that fail to capture the spatial variability of soils within polygons, particularly in complex or inaccessible terrains (Kempen et al., 2012). The static nature of traditional maps, coupled with the challenges of rapidly upscaling these methods to large areas, limits their utility in dynamic landscapes or extensive regions that require fine-scale soil information (Grunwald et al., 2011; Wadoux and McBratney, 2024). Modern approaches to soil series mapping leverage advancements in geospatial technology, remote sensing, and data analytics (Abdulraheem et al., 2023). Digital Soil Mapping (DSM), in particular, has revolutionized the field by enabling the prediction of soil properties and class across large areas with high spatial resolution (Garg et al., 2020). It integrates data from multiple sources, including satellite imagery, digital elevation models (DEMs), and climate datasets, to model relationships between soil properties and environmental covariates (Lamichhane et al., 2019; Fathololoumi et al., 2020). These models are often driven by machine learning algorithms, which excel at analysing complex and non-linear relationships in large datasets (Kalambukattu et al., 2018). DSM is cost-effective and time-efficient, requiring fewer field surveys to validate model predictions (Kempen et al., 2012). Furthermore, DSM is dynamic, capable of incorporating temporal changes in soil properties driven by factors such as climate change or land-use alterations.

Despite the rapid evolution of machine-learning-based DSM approaches in soil science, their application to detailed soil series delineation remains limited. Most global efforts-such as those by Hengl et al. (2017), Brungard et al. (2015), Heung et al. (2017), Zeraatpisheh et al. (2019), Taghizadeh-Mehrjardi et al. (2019), Reza et al. (2024), Chattaraj et al. (2025a), Chattaraj et al. (2025b) -have predominantly targeted the prediction of continuous soil properties, limited number of studies have attempted to delineate soil series or soil classes through DSM approaches (Esfandiarpour-Boroujeni et al., 2020; Mosleh et al., 2017; Zeraatpisheh et al., 2017), where predictions were made using field observations and environmental covariates, but the resulting maps were not quantitatively assessed for spatial agreement with large-scale TMU-based soil series maps.

Overall, accurate soil-class or type mapping, followed by the aggregation of similar soil series into *Soil Management Zones (SMZs)*, is foundational for watershed management and agricultural land-use planning vis-à-vis precision agriculture. Traditionally, detailed soil surveys have played a pivotal role in integrated watershed development. However, in many developing nations,especially in geographically expansive countries like India, these surveys have gradually declined, hindered by limited expertise, labour intensive, high operational costs, and substantial time requirements. This decline underscores the need for an alternative, dynamic methodology that can generate soil-class information down to the soil-series level. DSM presents a promising solution to these challenges. Despite its growing appeal, the practical effectiveness of DSM for detailed soil survey mapping remains underexplored. Evaluations of DSM's capability to serve as a viable, scalable alternative to *conventional* methodologies are notably lacking. The present study extends beyond soil attribute (S_a) mapping to develop a machine-learning-integrated DSM framework for operational, detailed soil survey through large-scale soil series mapping.

The Eastern Ghats of India is one of the oldest landmasses and biodiversity hotspots (Mandal et al., 2020) in the country, encompassing the states of Odisha, Andhra Pradesh, Telangana, Karnataka, and Tamil Nadu (Banerjee, 1990). In the Eastern Ghat Highlands, geomorphic versatility and geologic complexity make the physiography highly significant, endowed with diversified soils (Bandyopadhyay et al., 2025) and land-use systems. In this backdrop, Boipariguda Sub-Watershed Cluster has been selected, which typically represents a miniature of Eastern Ghat Highlands of Odisha, India. The region's geomorphic diversity, geological variability, and dynamic land-use patterns provide a suitable setting to evaluate the performance of DSM.

Building on these scientific gaps, this study adopts a dual-framework approach-integrating *conventional terrain-mapping unit (TMU)*-based soil series delineation using *conventional* detailed soil survey with a novel application of DSM. The objectives of the study are fourfold: (i) to delineate soil series using both *conventional* detailed soil survey at 1:8000 scale and DSM-based machine learning models at high resolution (5 m); (ii) to compare the predictive performance, spatial precision, and operational feasibility of the two methods; (iii) to identify and quantify key soil-forming factors influencing series-level spatial variability through model-based analysis; and (iv) to delineate *Soil Management Units (SMUs)* based on DSM method and assess its applicability for precision land management with reference to *conventional* approach.

2. Material and methods

2.1. Study area

The study was conducted in the Boipariguda Sub-Watershed Cluster, located in the Boipariguda block of Koraput district, Odisha, India. Geographically, it extends between 18°35′35″N to 18°47′32″N latitudes and 82°21′15″E to 82°30′35″E longitudes, covering an area of 20,722 ha (Fig. 1). This region falls within the Godavari River Basin, specifically in the Machhkund-Kolab catchment and the Upper Kolab sub-catchment, a part of the Kolab Saptadhara watershed (District Census Handbook, 2011; MWR, 2014). The Eastern Ghat Highlands are dominated by highly weathered red and lateritic soils (Sarkar et al., 2005) and are famous for their picturesque cultural landscapes under shifting cultivation by indigenous farmers, known as *Podu* farming (Orissa Review, 2006). The area exhibits complex geomorphology comprising dissected plateaus, rugged hills, and narrow valleys, with elevation ranging from 45 to 1141 m above mean sea level (Behera et al., 2024). Geologically, it is dominated by *granite-gneissic* and gneissic complex followed by *charnockite* and *Khondalite* formations (Geological Survey of India, 2002). It is situated in the sub-tropical and sub humid (moist) eco-sub region with deep loamy to clayey red and lateritic soils (Velayutham et al., 1999). The study area experiences warm to hot summers, cool winters, and a pronounced southwest monsoon (DDMP, 2017). The annual rainfall ranges between 1650 and 1700 mm, with around 80% falling during the monsoon season (District Irrigation Plan, 2023). Erratic rainfall patterns and a short growing period (about 11 weeks) make the area drought-prone (District Irrigation Plan, 2023). Only 30.9% of the land is used for cultivation, dominated by rainfed minor millets, while rice and vegetables are grown in limited irrigated areas (NABARD, 2022). These environmental and soil constraints underscore the need for high-resolution soil series mapping to inform agricultural planning and watershed development (Sahu and Mishra, 2005).

2.2. Conventional soil series mapping using Terrain Mapping Units (TMUs)

This study was conducted under the REWARD Odisha Programme, supported by the World Bank, using WorldView−2 imagery and a 5 m DEM generated and supplied by the Odisha Space Applications Centre, under the technical programme of the Government of Odisha. Land use, landforms and slope layers were generated at a working scale of 1:8000 and integrated to derive terrain mapping units following the REWARD Odisha Field Guide (Bandyopadhyay et al., 2022). These units

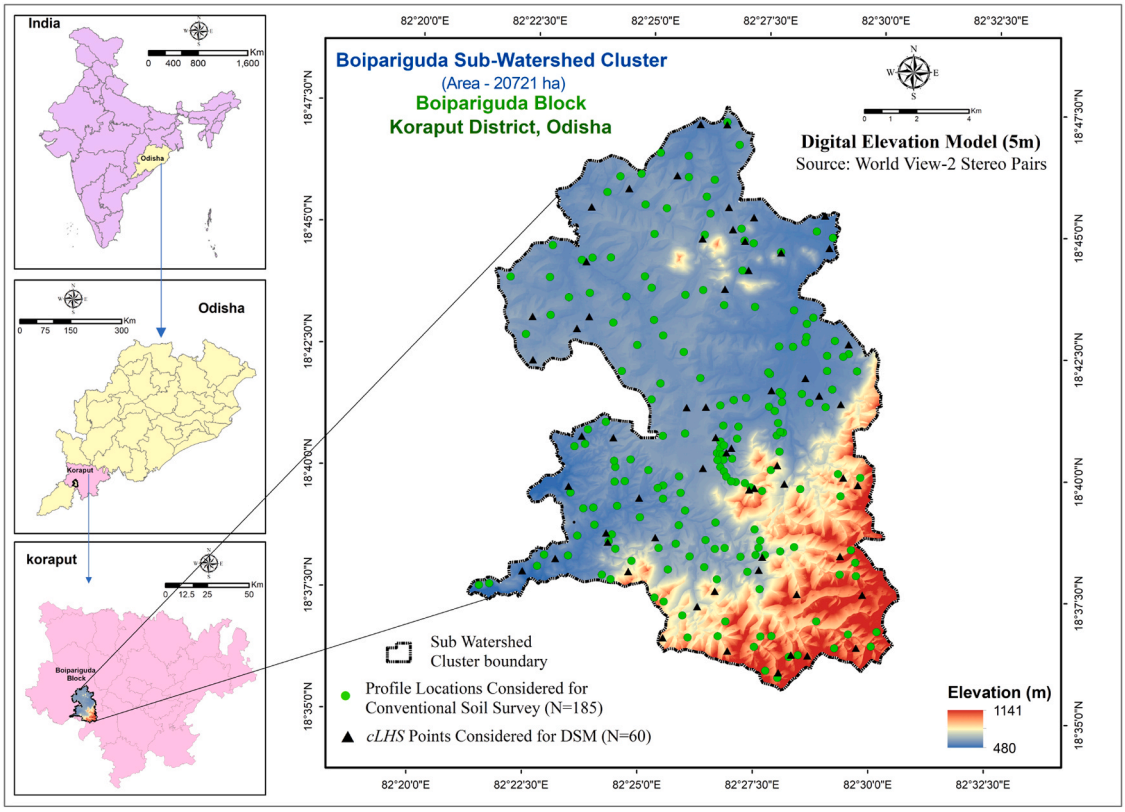


Fig. 1. Location of the study area. Green circles represent profile locations considered for *conventional* soil survey (N = 185), and black triangles denote *cLHS* points considered for digital soil class mapping (DSM, N = 60).

adequately captured soil-landform relationships and topographic variability required for delineation of soil series and soil management units. The adopted scale meets cartographic requirements, in which data resolution must be substantially finer than the minimum mappable unit. Ultra-large-scale mapping (e.g., 1:1250) would require sub-metre elevation data and a much denser field network, neither of which is operationally feasible or necessary for landscape-level soil pattern analysis. Therefore, the 5 m DEM represents an optimal compromise for digital soil mapping in the rugged hill landscapes of the Eastern Ghats.

In this study, *TMUs* formed the foundational framework (base map) for *conventional* soil series delineation, leveraging high-resolution geospatial data and systematic field observations. *TMUs* were derived by integrating geology, landform, slope, and land use/land cover information, following the Standard Operating Procedure (SOP) developed by ICAR-NBSS&LUP for large-scale soil mapping (Bandyopadhyay et al., 2022). Cloud-free World View 2.0 imagery (0.53 m) acquired for the dry season (November-March, 2022–23) served as the primary base to extract detailed Land Use/ Land Cover (LULC) and landform classes at 1:8000 scale. The slope component was generated from a stereo-derived World View 2.0 Digital Elevation Model (DEM) having 5 m spatial resolution (Source: Odisha Space Applications Centre (ORSAC), Govt. of India).

Topographical Sheets of Survey of India were used to delineate permanent LULC features like habitations, waterbodies, rivers, drainage streams, roads, railway tracts, etc. Legacy data of Geological Survey of India available at 1:50,000 scale was used to derive important geologic formations in the study area (https://geodataindia.gov.in). *TMUs* were then delineated by systematically interpreting major physiographic regions and geological formations, then differentiating landforms-such as hills, ridges, uplands, and valleys-within each geological unit based on land use, slope, contour intervals, and image characteristics, ultimately leading to a physiography-landform map at the sub-watershed scale (Bandyopadhyay et al., 2022). The *TMUs* legend followed a structured coding scheme comprising 4–5 letters: the first letter denoting geology, the next two representing landform, followed by slope and land use, respectively (Fig. 2).

The study was conducted in two distinct geological formations, viz., the charnockite and the granite-gneissic complex of the Koraput formations (Geological Survey of India, 2002). The present endeavour focuses on the influence of geology, landforms, and vegetation (land use-land cover) on the occurrence of varying soil types in the Sub-Watershed cluster. Intensive traversing has been carried out across all the *TMUs* in selected transects from ridges to valleys.

A total of 245 soil profiles were excavated to investigate soil morphological properties (Soil Survey Staff, 2003) across all TMUs in a stratified manner (Fig. 1), taking into account the catenary sequence and targeting the dominant soil series within each unit to ensure adequate representation of pedogenic variability. Horizon-wise soil samples were collected from all profiles to a maximum depth of 200 cm or to lithic/paralithic contact, where encountered. The samples were pre-processed and analysed for key physicochemical properties using standard laboratory procedures (Sparks, 1996). Of these, 185 profiles were used for conventional soil series mapping, while the remaining 60 profiles were reserved for calibrating DSM-based soil series prediction. Details regarding the subset used for DSM model calibration and validation are provided in Section 2.3.3.

Subsequently, soil series were delineated by correlating observed pedogenic characteristics with *TMUs*-specific geology, landform, slope, and land use attributes, ensuring taxonomic consistency and spatial coherence in capturing soil variability across the landscape (Bandyopadhyay et al., 2022). The criteria for defining soil series and their taxonomic placement at the subgroup level were based on USDA Soil Taxonomy and harmonized with the legacy classification framework of ICAR-NBSS&LUP, particularly the approaches of Sehgal (1986), Soil Series of Odisha (Sarkar et al., 2005), and the REWARD Odisha Field Guide (Bandyopadhyay et al., 2022; Singh et al., 2016). Soil series

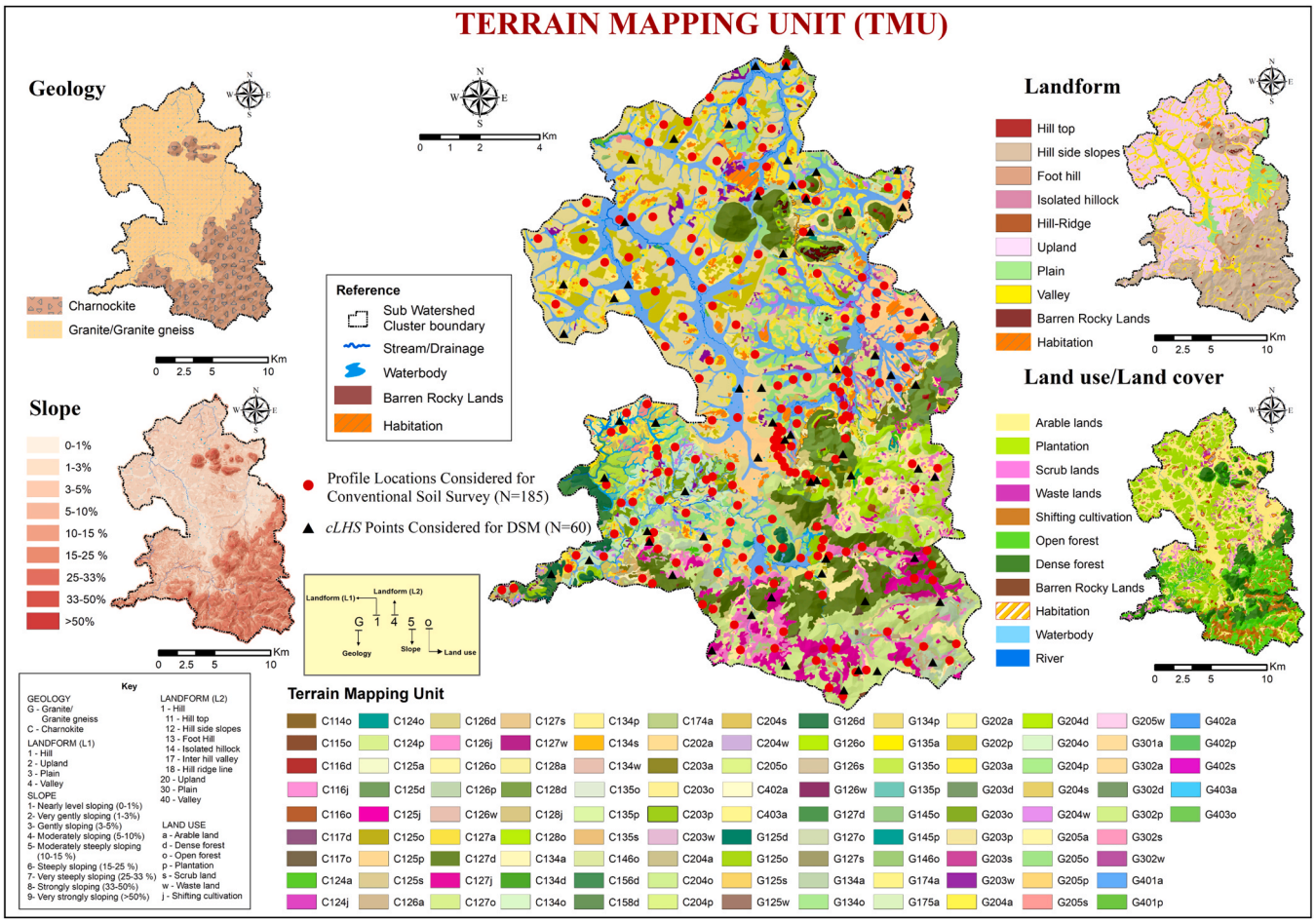


Fig. 2. *Terrain Mapping Units (TMUs)* delineated based on integrated analysis of geology, slope, landform, and land use/land cover for detailed soil resource inventory and digital soil mapping in the study area.

differentiation primarily considered parent material and geology, landform, effective soil depth, diagnostic horizon characteristics, texture, matrix colour, drainage, gravelliness and selected physico-chemical properties, including soil pH. Locally recognized soil series names were retained to facilitate regional applicability, while their taxonomic equivalence was established by assigning both USDA Soil Taxonomy and WRB classifications at the subgroup level.

2.3. Digital Soil Mapping (DSM) of Soil Series Using a Machine Learning Framework

2.3.1. Environmental covariates

Based on the SCORPAN approach, four broad soil-forming factors namely climate, satellite (proxy for organisms' factor), terrain, and geology (parent material), were incorporated as covariates to predict soil series vis-à-vis soil management zones. Among these factors, *TMU*, LULC, and landform were included, with their delineation methods detailed earlier in Section 2.2. Besides, World View 2.0 DEM of 5 m resolution were analysed in Saga-GISv8.4.0 for deriving primary and secondary terrain attributes like elevation, slope, channel network base level (CNBL), channel network distance (CND), relative slope position (RSP), valley depth (VD), multi-resolution ridge top flatness (MRRTF), multi-resolution index of valley bottom flatness (MRVBF), Slope Length and Steepness Factor (LS), and Topographic Wetness Index (TWI). Satellite imagery from World View 2.0 (spatial resolution of 0.53 m) was processed to derive vegetation indices such as the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Soil Adjusted Vegetation Index (SAVI), which were incorporated as proxies for net primary productivity and crop cover dynamics (Pettorelli et al., 2005). Thirty-year mean climate data were obtained from the WorldClim database and processed to extract key variables, including mean annual temperature (MAT), maximum temperature (T_{max}), minimum temperature (T_{min}), and mean annual precipitation (MAP). Additionally, information on the underlying geological composition of the parent material was incorporated using 1:50,000 scale maps from the Geological Survey of India (2002) (Table 1).

2.3.2. Variable Selection and Soil-Landscape Relation Pathway Modelling

To refine soil series prediction using the DSM framework, both variable selection and process pathway analysis were integrated into the workflow. This ensured that only the most influential and non-redundant covariates were used in the final model.

The *Boruta* algorithm, a wrapper method built around the *Random Forest classifier* (Kursa and Rudnicki, 2010), was used to identify all relevant covariates contributing to soil series differentiation. By comparing the importance of each actual variable against a set of randomized shadow variables, *Boruta* ranks predictors based on their statistical relevance. It not only identifies the variables with the highest predictive relevance but also eliminates weakly contributing variables.

In parallel, to quantify the hierarchical controls of soil-forming factors on soil series occurrence, a Bayesian structural equation modelling (SEM) framework was implemented by integrating latent variable construction with a multinomial logistic regression structure, thereby accommodating the categorical (nominal) nature of the response variable. Four conceptual latent constructs-Relief (R), Climate (C), Organism (O), and Parent Material (P)-were defined following the soil-forming factor

Table 1
Description environmental covariates used as inputs in model.

Types	Variables	Description	Source
Indices	Rabi_NDVI	Normalized Difference Vegetation Index of winter season	World View 2.0 (0.53 m)
	Rabi_SAVI	Soil Adjusted Vegetation Index of winter season	
	Rabi_EVI	Enhanced Vegetation Index of winter season	
Terrain	CNBL	Channel Network Base Level	World View 2.0 DEM (1 m)
	CND	Channel Network Distance	
	RSP	Relative Slope Position	
	Elevation	–	
	Slope	–	
	LS	Slope length and steepness	
	VD	Valley depth	
	TWI	Topographic Wetness Index	
	MRRTF	Multi-Resolution Ridge Top Flatness	
	MRVBF	Multi-Resolution Index of Valley Bottom Flatness	
Climate	MAT	Mean Annual Temperature	Worldclim data
	MAP	Mean Annual Precipitation	
	Tmax	maximum temperature (°C)	
	Tmin	minimum temperature (°C)	
Geology	Geology	Parents Materials	bhukosh.gsi.gov.in

paradigm (McBratney et al., 2003), and were operationalized using observed environmental covariates. Specifically, Relief was represented by CNBL, elevation, slope, VD, CND, MRVBF, and landform; Climate by MAP and MAT; and Organism by NDVI and LULC, while Parent Material was directly represented by geology.

Because these covariates are not directly observable, latent scores were derived using a dimension-reduction approach based on principal component analysis (PCA), retaining the first component in each group to represent the dominant variance structure. These latent variables were subsequently standardized and used as predictors in a Bayesian multinomial logistic regression, in which soil series (categorical with multiple unordered classes) was specified as the dependent variable. The probabilistic formulation assumes that the likelihood of each soil class follows a categorical distribution with class probabilities modelled through a softmax link function of the latent predictors.

Weakly informative Gaussian priors were assigned to regression coefficients to stabilize estimation under limited sample size and to avoid overfitting, consistent with recommendations for Bayesian generalized linear models (Gelman et al., 2014). Posterior distributions of model parameters were estimated using Markov Chain Monte Carlo (MCMC) sampling, and convergence diagnostics (trace behavior and effective sample size) were assessed to ensure stable inference. The resulting posterior means of regression coefficients were interpreted as standardized pathway effects linking latent soil-forming factors to soil class probabilities. In parallel, interrelationships among latent constructs were quantified using Pearson correlation coefficients, representing covariance pathways within the Bayesian framework. The overall modelling strategy thus represents a Bayesian SEM-type pathway analysis, in which the measurement component (latent variable construction) and structural component (multinomial regression) are explicitly linked, enabling simultaneous interpretation of factor interdependence and their probabilistic influence on soil series distribution.

2.3.3. Random Forest Model Development and Soil Series Prediction

The *Random Forest (RF)* algorithm (Breiman, 2001) was employed as the core classification model due to its robustness in handling high-dimensional, non-linear relationships, and categorical response variables. In this study, out of the 245 soil profiles, 60 representative profiles were selected using the *Conditioned Latin Hypercube Sampling (cLHS)* algorithm (Minasny and McBratney, 2006) to calibrate the RF classification model. The remaining 185 profiles were retained as an independent dataset for subsequent validation of the DSM-derived soil series map. The cLHS algorithm was adopted to optimize the calibration dataset by maximizing coverage of the multidimensional environmental covariate space. Unlike conventional physiography- or traverse-based sampling approaches, which often produce spatially clustered and information-redundant observations, cLHS explicitly targets representative as well as rare combinations of environmental conditions. Consequently, although only 60 profiles were used for model calibration, the selected dataset adequately captured the environmental variability governing soil formation and spatial differentiation across the study area, thereby reducing sampling bias and minimizing the need for model extrapolation (Padarian et al., 2019; Dai et al., 2019).

These soil profile observations, integrated with the set of environmental covariates (Section 2.3.1), were standardized and spatially aligned to ensure consistency, and subsequently employed to generate a high-resolution (5 m) soil series map-demonstrating the practical potential of machine learning in detailed soil survey. The calibration (training) dataset used for RF modelling is represented as follows:

$$D = \{(x_i, y_i)\}_{i=1}^{N}, x_i \in \mathbb{R}^p, y_i \quad \in \{1, \quad 2, \ldots..., \quad K\} \tag{1}$$

Where, where x_i is a vector of p predictor variables (covariates) and y_i is the soil class label among K possible classes.

Based on this calibration dataset, a random forest constructs T classification trees *{ht(x), t = 1, 2, …,T},* where each tree is trained on a bootstrap sample of D, and at each node, a random subset of predictors is used for splitting.

The final classification decision for a new input x is made by majority voting:

$$\hat{y}(x) = arg_{k\in\{1,\ldots..,K\}} \max \sum_{t=1}^{T} 1(h_t(x) = k) \tag{2}$$

Where, h_t *(x)* = predicted class from the tth decision tree, *1(·)* = indicator function (1 if true, 0 otherwise), and $\hat{y}$*(x)* = predicted soil class for input x.

In the present work, the *RF classification model* was used in R package randomForest and terra with parameters set at 2000 *ntrees* and an *mtry* value of 6. A grid search with 10-fold cross-validation was applied to identify the optimal parameter combination that minimized classification error (OOB error rate). The caret package in R was utilized for systematic parameter optimization.

2.4. Uncertainty quantification

Uncertainty associated with predicted soil class assignments was quantified using class-membership probabilities generated by the Random Forest model. Two complementary uncertainty metrics were computed to assess the reliability of soil class predictions.

Entropy-based uncertainty: Shannon entropy (Kempen et al., 2009) was used to quantify classification ambiguity following the equation below.

$$H = -\sum_{k=1}^{K} Pk\log(Pk) \tag{3}$$

Where, Pk represents the class-membership probability of a pixel belonging to soil class k. Higher entropy values indicate greater uncertainty in assigning a pixel to a specific soil class, reflecting increased ambiguity among competing classes.

Classification confidence: Classification confidence was also computed following,

$$C = \max(Pk) \tag{4}$$

.

where, where Pk represents the class-membership probability of a pixel belonging to soil class k. Larger values of C indicate higher

confidence (low uncertainty) in the predicted class assignment. Spatial uncertainty maps were generated to highlight areas where soil class predictions are less reliable, thereby indicating locations where additional field sampling or validation may be required.

2.5. Delineation of Soil Management Units (SMUs)

Soils exhibit pronounced spatial variability arising from diverse soil-forming factors and pedogenic processes. This heterogeneity in soil properties leads to differential patterns of moisture retention, nutrient availability, and consequently, variations in crop growth and the efficiency of externally applied inputs. Implementing variable-rate input applications therefore necessitates the delineation of land areas-whether at field, regional, or watershed scales-into homogeneous *SMUs* to optimize productivity while mitigating adverse environmental impacts (Ramamurthy et al., 2016). The central concept of *SMUs* lies in regrouping similar soil series that inherently bear similar landform, slope, and land-use systems, bounded by TMUs, to obtain a manageable number of mapping units for land-use planning interventions (Ramamurthy et al., 2016; Ghosh et al., 2018; Reza et al., 2020). In the present study we hypothesized that internal soil drainage, soil depth, texture of soil control section (0–100 cm), soil reaction (acidity/alkalinity) and soil erosion status were the major limiting characteristics of soils that affect the changes in land use systems. As a result, these characteristics were considered critical indicators for obtaining *SMUs*.

2.6. Accuracy Assessment of DSM-based Soil series and SMUs mapping

To evaluate the classification agreement between traditional and DSM-based approaches, Cohen's Kappa statistic (Cohen, 1960) was employed to quantify the agreement between DSM predictions and conventional mapping, as it accounts for chance agreement and provides a robust measure of categorical consistency, where values close to 1 indicate almost perfect agreement, 0.61–0.80 substantial, 0.41–0.60 moderate, and values near 0 minimal agreement beyond randomness (Landis and Koch, 1977). Accuracy was assessed at two levels: point-based validation against 185 independent soil profiles withheld for testing, which provided field-level reliability and raster-based pixel-to-pixel comparison across the entire spatial domain, which captured spatial congruence at map scale.

3. Results

3.1. Soil Series mapping using Conventional approach

3.1.1. Soil-TMU relationship in depicting soil series mapping

Representative soil-TMU relationships in the two major geological formations, namely charnockite and granite-gneissic complexes, are illustrated in Figs. 3 and 4, while detailed morphological characteristics of the representative soil series are presented in Table 2. In the charnockite landscape, soil development followed a distinct topographic sequence from hillslopes (>650 m) through inter-hill valleys, foothills, uplands and valleys (630–640 m). Hillslope soils occurring on steep (15–25%) to very steep (>33%) slopes under scrubland and open forest were moderately deep to deep (75–150 cm), slightly to moderately gravelly, strongly acidic (pH 5.0–5.5) and well-drained, with A-Bt-BC horizonation. In contrast, foothill and upland positions supported very deep (>150 cm), well-drained soils with well-developed Bt horizons, clay loam to clayey textures and red lateritic characteristics, indicating comparatively advanced pedogenic development under more stable geomorphic conditions. Valley soils occurring on gentle (1–5%) slopes under paddy-fallow systems were consistently very deep (>150 cm), non-gravelly, slightly acidic to neutral (pH 6.0–7.3), with silty clay loam to silty clay textures and prominent Bg horizons, reflecting seasonal waterlogging and gleyization.

A similar toposequence-controlled distribution was observed in the granite-gneissic landscape, extending from isolated hillocks (>750 m) to foothills, inter-hill valleys, uplands, plains and valleys (600–625 m). However, soils developed over granite-gneiss were generally shallower (50–75 cm) at higher elevations, severely gravelly, strongly acidic and exhibited relatively weaker profile development with A-Bw-C horizon sequences. The foothill and upland soils were deeper with argillic horizon development, whereas plains and valleys contained very deep, fine-textured, imperfectly to poorly drained soils with Btg/Bg horizons that were morphologically comparable to those observed in the charnockite valleys.

Eventually, both geological formations exhibited a systematic soil-landscape relationship, with hillslope soils characterized by shallow depth, coarse texture, high gravel content, strong acidity and good drainage, foothill and upland soils by increased profile depth, finer textures and greater pedogenic development, and valley soils by very deep profiles, finer textures, poor drainage and gleyic features. Nevertheless, the charnockite landscape consistently exhibited greater soil profile maturity, as evidenced by increased profile depth, stronger matrix colour development, and progressive clay accumulation along the toposequence, whereas the granite-gneissic terrain retained comparatively less-developed profiles at higher elevations.

3.1.2. Distribution of soil series

The *conventional* soil classification based on *TMU*s identified 22 soil series across the Boipariguda sub-watershed in the Eastern Ghats, Odisha, reflecting the complex interactions among lithology, topography, vegetation, and land-use systems. Among these, the most spatially dominant series was BPG-A, covering about 4661 ha of the total geographical area of 20722 ha, predominantly located in the southern, south-eastern, and central hilly forested regions under *Charnokite* geology (Fig. 5a). This series is characterized by moderately deep, strongly acidic, red loamy soils with a moderate gravel content, occurring on hillside slopes with mostly under forest cover. The south-central, northern, and western uplands under *Granite-gneissic* geology were extensively covered by BPG-N (4759 ha) and BPG-M (1882 ha), both representing very deep, well-drained, strongly acidic soils under plantation and arable land uses. The valley bottoms and lowland plains hosting major agricultural land across the region were marked by BPG-R (3059 ha) and BPG-P (743 ha), both comprising very deep, poorly drained alluvial soils with high clay content, indicative of depositional zones under intensive cultivation. Less extensive soil series, such as BPG-E (15 ha), BPG-U (32 ha), and BPG-G (19 ha), were restricted to isolated hillocks and fragmented forest patches, often in degraded or topographically extreme areas. A comprehensive description and spatial distribution of all *conventionally* identified soil series are provided in Table 2.

3.2. Soil Series Mapping Using ML-based DSM approach

3.2.1. Deciphering Factors affecting Spatial Occurrences of Soil Series

To unravel the hidden drivers of soil series distribution, the analysis began with *Boruta* feature selection, which served as a filter to separate the truly influential covariates from noise. Only those variables with an importance score above the threshold value of 5 survived as "confirmed" predictors, while the rest were discarded. These refined predictors then set the stage for Structural Equation Modelling, enabling a deeper exploration of the direct and indirect pathways shaping soil series occurrences.

3.2.1.1. Variable Importance and Environmental Controls on Soil Series Prediction. The *RF classification model*, coupled with *Boruta* based feature selection, identified TMU as the highest-scoring categorical predictor (mean importance score of 20.52), followed by landform (16.03) and land use/land cover (14.19) (Fig. 6a) Other important

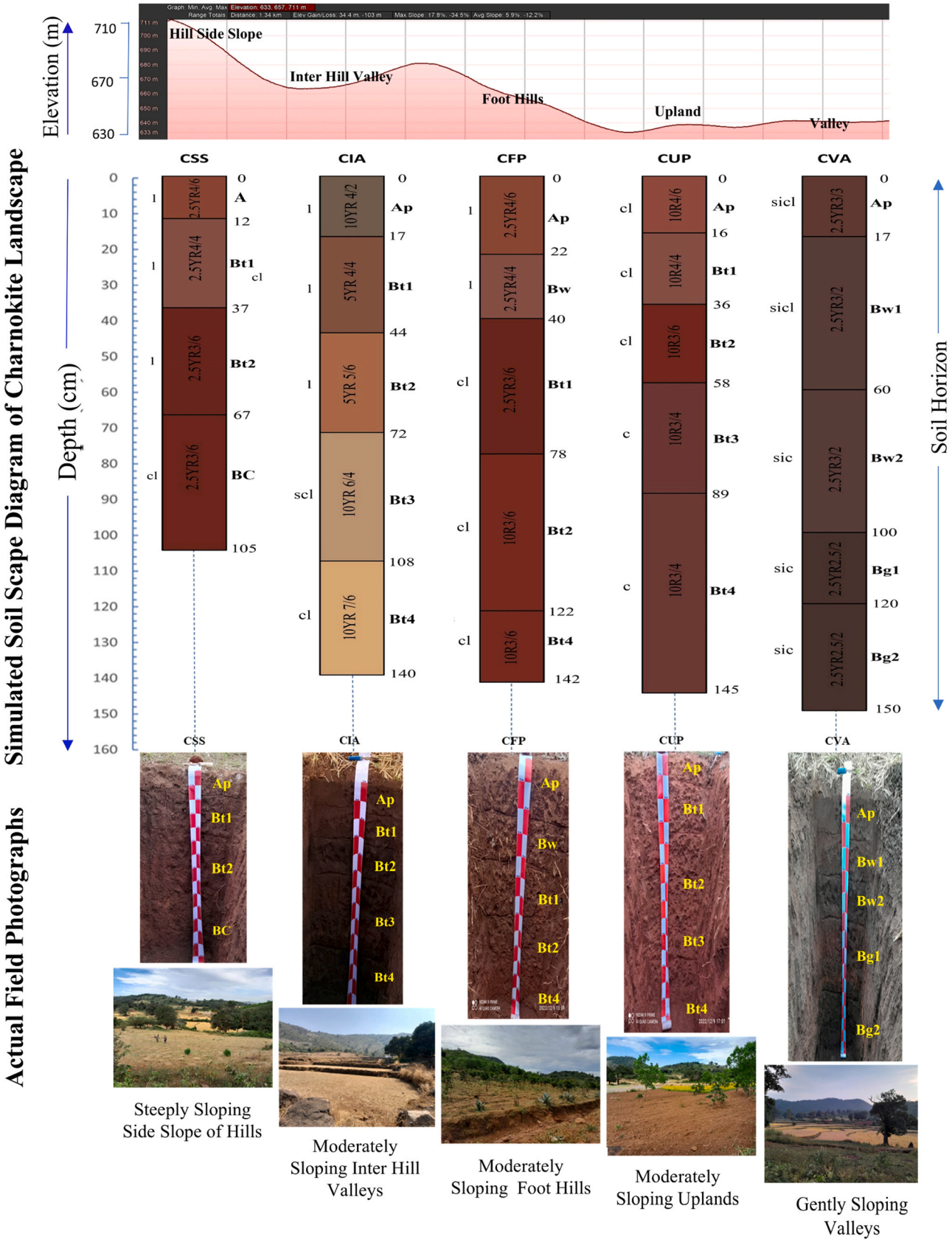


Fig. 3. Soil-TMU relationship diagram in Charnockite landscape of the study area. Abbreviations: CSS- Side slope of Hills (Charnockite) under Srublands; CIA- Inter-hill Valleys (Charnockite) under Agriculture; CFP- Foot Hills (Charnockite) under Plantations; CUP- Uplands (Charnockite) under Plantations; CVA- Valleys (Charnockite) under Agriculture.

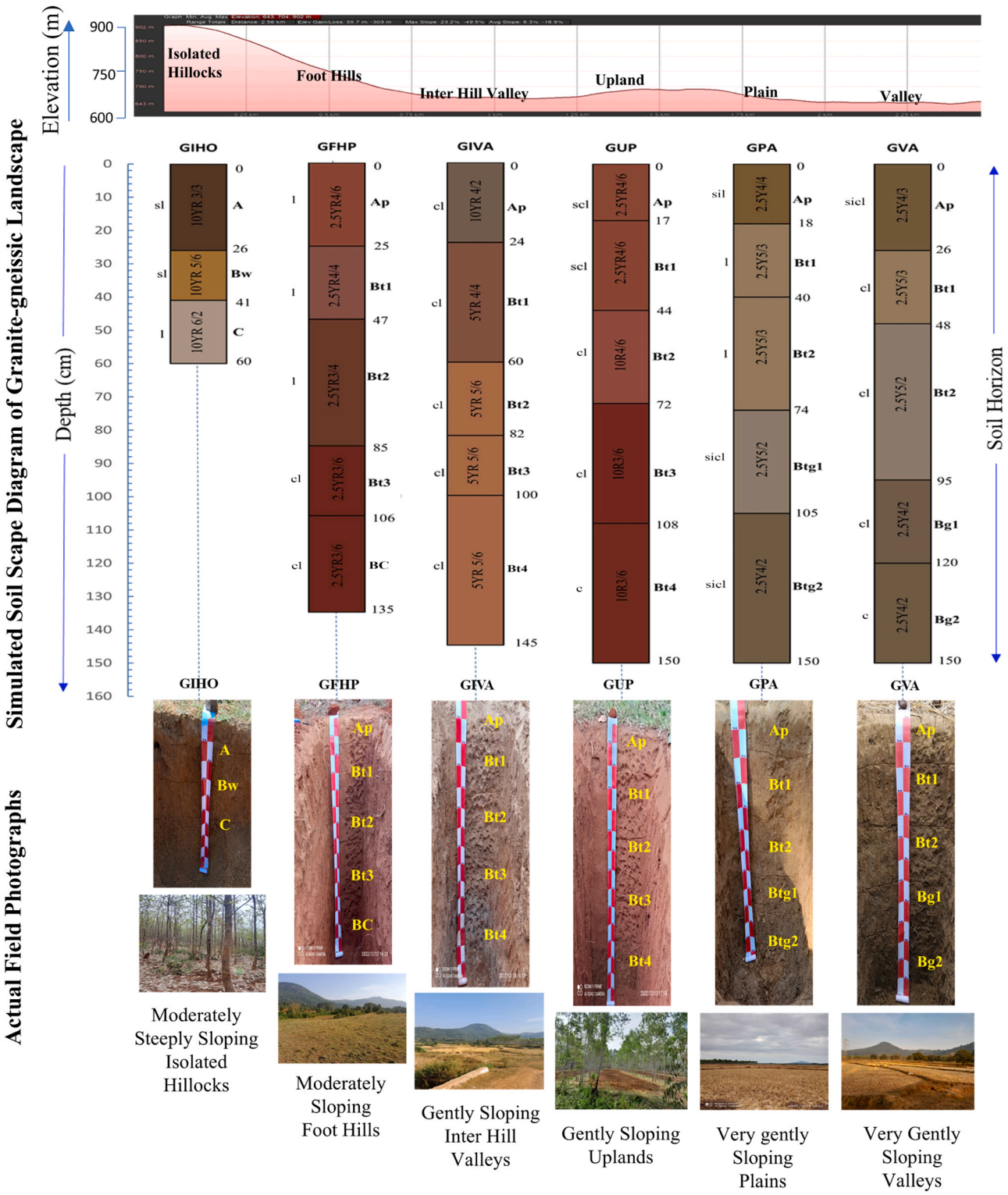


Fig. 4. Soil-TMU relationship diagram in Granite-gneissic landscape of the study area. Abbreviations: GIHO- isolated Hillocks (Granite-gneissic) under Open Forests; GFHP- Foot Hills (Granite-gneissic) under Plantations; GIVA- Inter-hill Valleys (Granite-gneissic) under Agriculture; GUP- Uplands (Granite-gneissic) under Plantations; GPA- Plains (Granite-gneissic) under Agriculture; GVA- Valleys (Granite-gneissic) under Agriculture.

predictors included MRVBF (10.56), geology (10.05), elevation (7.48), and vegetation indices such as Rabi-season NDVI, SAVI, and EVI (all above 7), indicating that subtle variations in terrain and seasonal vegetation dynamics also contribute significantly to soil variability. Hydrological and climatic covariates such as CNBL, CND, slope, TWI, mean annual precipitation (MAP), mean annual temperature (MAT), and maximum temperature (Tmax) showed moderate importance (scores between 5.0 and 7.0), suggesting a complementary role in soil formation processes. The consistent importance scores of the selected features underscore the multidimensional nature of soil series distribution, shaped by a complex interplay among terrain, climate, vegetation, and land-use factors. This aspect will be further elaborated in the

Table 2
Distribution of Terrain Mapping Units (TMUs) based soil series of Boipariguda sub watershed in different geological set up.

Geology	Landform	TMU Code	Soil Series	Area (ha)	Description	USDA Soil Taxonomy (WRB)
Charnockite	Hill top Hill side slopes	C115o C116d C127d C127o	BPG-A	4661	Moderately deep, somewhat excessively drained, moderately gravelly, strongly acidic red and lateritic soils with loam to sandy clay loam texture, occurring on hillside slopes with forest under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
		C126a C126p	BPG-B	446	Deep, somewhat excessively drained, slightly gravelly, moderately acidic, red, and lateritic soils with loam to sandy clay loam texture occurring on hillside slope with arable land with intrusion of plantation and tree clad area under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
		C124a C124p C125a C125p	BPG-C	111	Deep, well-drained, non-gravelly, moderately acidic, red, and lateritic soils with loam to clay loam texture occurring on hillside slopes with arable land with intrusion of plantation and tree clad areas under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
	Hill top Hill side slopes	C116j C126j C127j	BPG-D	1286	Deep, somewhat excessively drained, slightly gravelly, moderately acidic, red, and lateritic soils with sandy clay loam to clay loam texture occurring on hillside slopes with shifting cultivation under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
		C125s C127w	BPG-E	15	Moderately-deep, somewhat excessively drained, moderately gravelly, very strongly acidic, red, and lateritic soils with sandy clay loam to clay loam texture, occurring on hillside slopes with scrub land and waste land under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
	Foot hill	C134a C134p	BPG-F	320	Very Deep, well-drained, slightly gravelly, strongly acidic, red and lateritic soils with clay loam to sandy clay loam texture occurring on the foothill with arable land with intrusion of forest, waste land, scrub land and tree class area under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
	Hill-Ridge	C156d C158d	BPG-G	19	Moderately shallow, somewhat excessively drained, severely gravelly, strongly acidic red loamy soils with loam to sandy clay loam texture, occurring on the hill ridge with forest under Charnockite landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
	Isolated hillock	C146o	BPG-H	7	Moderately shallow, somewhat excessively drained, severely gravelly, very strongly acidic red loamy soils with loam to clay loam texture, occurring on the isolated hillocks with forest and tree clad area under Charnockite landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
	Inter hill valley	C174a	BPG-I	204	Very Deep, moderately well-drained, non-gravelly, moderately acidic, red and yellow soils with loam to clay loam texture occurring on inter-hill valley of arable land under Charnockite landscape.	Fine-loamy Oxyaquic Haplustalfs (Stagnic Luvisols)
	Upland	C203a C204a C204p	BPG-J	175	Very Deep, well-drained, non-gravelly, strongly acidic, red and lateritic soils with loam to clay loam texture occurring on the upland with arable land, plantation, forest, waste land and scrub land under Charnockite landscape.	(Fine Rhodic Paleustalfs) (Rhodic Nitisols) (Fine Rhodic Paleustalfs) (Rhodic Nitisols)
	Valley	C402a C403a	BPG-K	127	Very Deep, somewhat poorly drained, non-gravelly, slightly acidic to neutral, alluvial soils with silty clay loam to silty clay texture occurring on valley with arable land under Charnockite landscape.	Fine Typic Endoaqualfs (Stagnic Gleysols)
Granite-gneissic	Hill side slopes	G125o G126d G127d	BPG-L	424	Moderately-deep, somewhat excessively drained, moderately gravelly, strongly acidic red loamy soils with sandy loam to loam texture occurring on hillside slopes with forest, scrubland and waste land under granite-gneissic landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
	Upland	G202a G203a G204a	BPG-M	1882	Very Deep, moderately well drained, non-gravelly, moderately acidic, red loamy and red and lateritic soils with clay loam texture occurring on the upland with arable land under Granite-gneissic landscape.	Fine Rhodic Paleustalfs (Rhodic Nitisols)
		G203p G204p	BPG-N	4759	Very Deep, well-drained, non-gravelly, strongly acidic, red and lateritic soils with clay loam to clay texture occurring on the upland with plantation and tree clad area under Granite-gneissic landscape.	Fine Rhodic Paleustalfs (Rhodic Nitisols)
		G203w G204o G204s	BPG-O	1297	Very Deep, well drained, slightly gravelly, strongly acidic, red and lateritic soils with clay loam to sandy clay texture occurring on the upland with scrub land, forest and waste land under Granite-gneissic landscape.	Fine Rhodic Paleustalfs (Rhodic Nitisols)
	Plain	G301a G302a	BPG-P	743	Very Deep, somewhat poorly drained, non-gravelly, slightly acidic, red and yellow soils with silty clay loam to silty clay texture occurring on the plain with arable land under Granite-gneissic landscape.	Fine-silty Oxyaquic Haplustalfs (Stagnic Luvisols)
		G302p G302s G302w	BPG-Q	173	Very Deep, moderately well drained, non-gravelly, moderately acidic, red loamy and red and yellow soils with clay loam to silty clay loam texture occurring on the plain with plantation, forest, scrub land and waste land under Granite-gneissic landscape.	Fine-silty Oxyaquic Haplustalfs (Stagnic Luvisols)
	valley	G401a G402a G403a	BPG-R	3059	Very Deep, poorly drained, non-gravelly, slightly acidic to neutral, alluvial soils with silty clay to clay texture occurring in the valley with arable land under Granite-gneissic landscape.	Fine Typic Endoaqualfs (Stagnic Gleysols)
	Foot hill	G134a G134p	BPG-S	160	Very deep, well-drained, slightly gravelly, strongly acidic, red and lateritic soils with loam to clay loam texture, occurring on the foothill with arable land under Granite-gneissic landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
		G134o G135o	BPG-T	163	Very Deep, somewhat excessively drained, moderately gravelly, very strongly acidic, red and lateritic soils with loam to clay loam texture, occurring on the foothill with forest and tree clad area under Granite-gneissic landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
	Isolated hillock	G146o	BPG-U	32	Moderately shallow, somewhat excessively drained, severely gravelly, strongly acidic, red loamy soils with sandy loam to loam texture, occurring on the isolated hillock with forest and tree clad area under Granite-gneissic landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
	Inter hill valley	G174a	BPG-V	37	Very Deep, moderately well-drained, non-gravelly, moderately acidic, red and yellow soils with loam to clay loam texture occurring on inter-hill valley with arable land under Granite-gneissic landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
Miscellaneous				629		
Grand Total				**20722**		

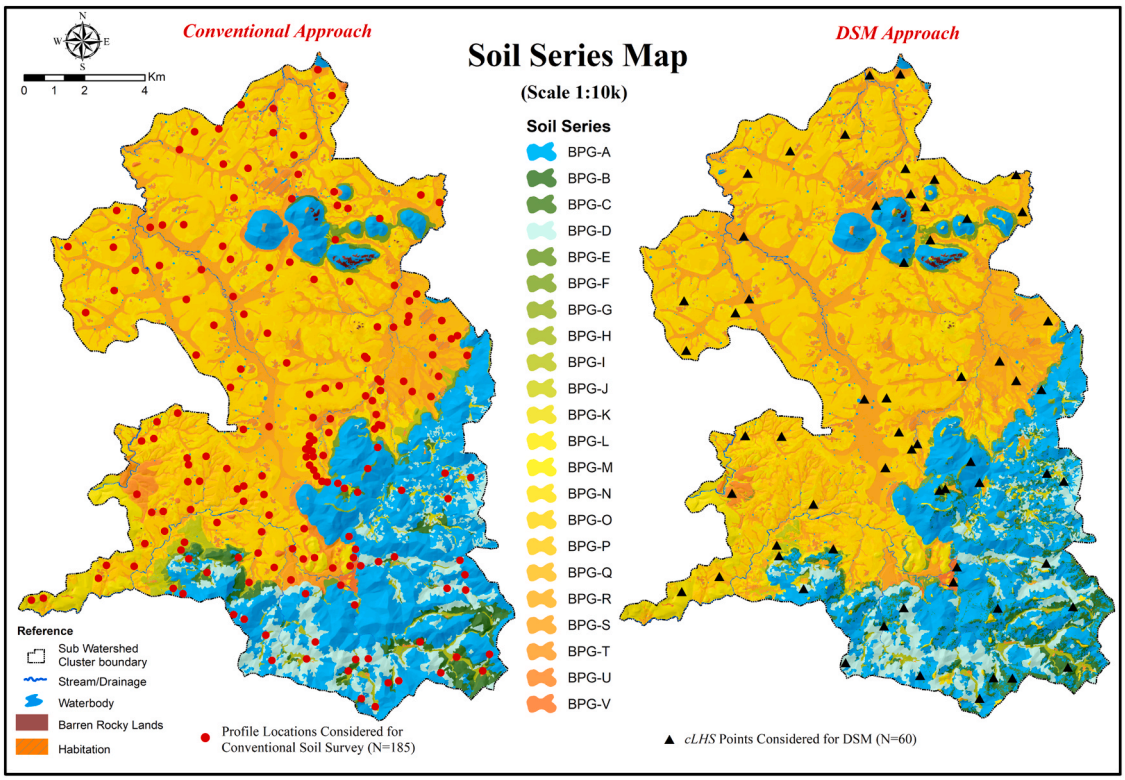


Fig. 5. Comparative soil series maps generated using the conventional (left) and DSM approach (right) at 1:10k scale, highlighting spatial variations in classification.

following Section.

3.2.1.2. Bayesian Structural Equation Modelling (SEM): Capturing Process Pathways of Soil Series occurrences. The Bayesian SEM framework revealed that among the broad soil-forming factors, Relief (R) exerted the strongest direct influence on soil series occurrence ($B = 0.80$), highlighting the dominant control of terrain attributes on pedogenesis. Among its observed indicators, elevation ($B = 0.74$) and CNBL ($B = 0.66$) showed strong positive contributions, followed by VD ($B = 0.11$), CND ($B = 0.08$), and slope ($B = 0.06$), while MRVBF and landform ($B = -0.01$) exhibited negligible to weak negative effects. Climatic factors (C) showed a moderate but negative direct effect on soil series ($B = -0.40$), with MAP contributing strongly ($B = 1.01$), whereas MAT had a negligible positive effect ($B = 0.02$). Organism (O) exerted a moderate positive influence ($B = 0.30$), driven primarily by LULC ($B = 1.01$), while NDVI showed a very weak negative contribution ($B = -0.03$). Parent material (P) also contributed positively ($B = 0.20$), with geology showing a strong direct loading ($B = 1.00$), confirming its foundational role.

The inter-factor relationships further illustrate the coupled nature of pedogenic controls. Relief showed a strong positive association with parent material (R ↔ P: $B = 0.757$) and a weak positive relationship with organism (R ↔ O: $B = 0.147$), while exhibiting a moderate negative relationship with climate (R ↔ C: $B = -0.614$). Climate, in turn, was weakly negatively associated with organism (C ↔ O: $B = -0.064$) and moderately negatively related to parent material (C ↔ P: $B = -0.529$). A weak positive relationship was also observed between organism and parent material (O ↔ P: $B = 0.080$). These interactions indicate that although Relief remains the dominant direct driver, its interplay with parent material, along with the weaker but notable linkages among climate and organism, collectively governs the spatial variability of soil series (Fig. 6b).

3.2.2. Spatial Prediction by RF classification model

The application of the *RF classification model* for DSM in the Boipariguda sub-watershed yielded highly promising results, reinforcing its potential as a reliable alternative to *conventional* soil series mapping approach. A total of 22 soil series originally identified through the conventional approach were also captured by the *RF classification model.* Spatial extent of the majority of these soil series was predicted with high accuracy under the RF-based DSM framework with reference to the *conventional* one (Fig. 5).

Ten major soil series having larger spatial extent demonstrated high agreement (<25% difference in mapped area) between the *conventional* and DSM-derived spatial extents, with minimal areal mismatches. These include BPG-D, BPG-M, BPG-I, BPG-K, BPG-N, BPG-V, BPG-A, BPG-O, BPG-R, and BPG-F. For instance, BPG-D, which occupies 1285.9 ha under the *conventional* approach, was predicted to cover 1276.6 ha by the RF model, resulting in a deviation of only −9.3 ha. BPG-M showed a similarly close match, with a *conventionally* delineated area of 1881.5 ha against a predicted area of 1874.7 ha. Other major series, such as BPG-I (204.0 ha vs. 196.7 ha), BPG-K (126.9 ha vs. 136.1 ha), and BPG-N (4758.6 ha vs. 4595.7 ha), also showed good agreement. Even in BPG-A-the most extensive unit in the study area-the RF model produced a close prediction, with 4127 ha mapped against 4660 ha in the *conventional* survey. Within this group, BPG-O, which spans 1296.6 ha in the *conventional* delineation, was slightly underrepresented in the RF output at 1013.5 ha. Five soil series showed moderate agreement, with moderate mismatches between the predicted and *conventional* maps. These series, while not as precisely matched as those in the high-agreement group, still retained a clear spatial structure in the DSM output, indicating reasonable predictive performance despite some misclassification at the edges or transition zones. The remaining six soil series exhibited a higher disagreement, where the RF-predicted areas either significantly exceeded or fell short of the *conventionally* mapped extents. Notably, BPG-B was overestimated by the model, with its area increasing from 446.2 ha to 833.3 ha. A similar trend was observed in BPG-G (19.0 ha to 51.8 ha), and BPG-L (420.1 ha to 928.2 ha, suggesting model confusion with adjacent series or overgeneralization in similar terrain contexts (Fig. 7). On the other hand, some series were markedly

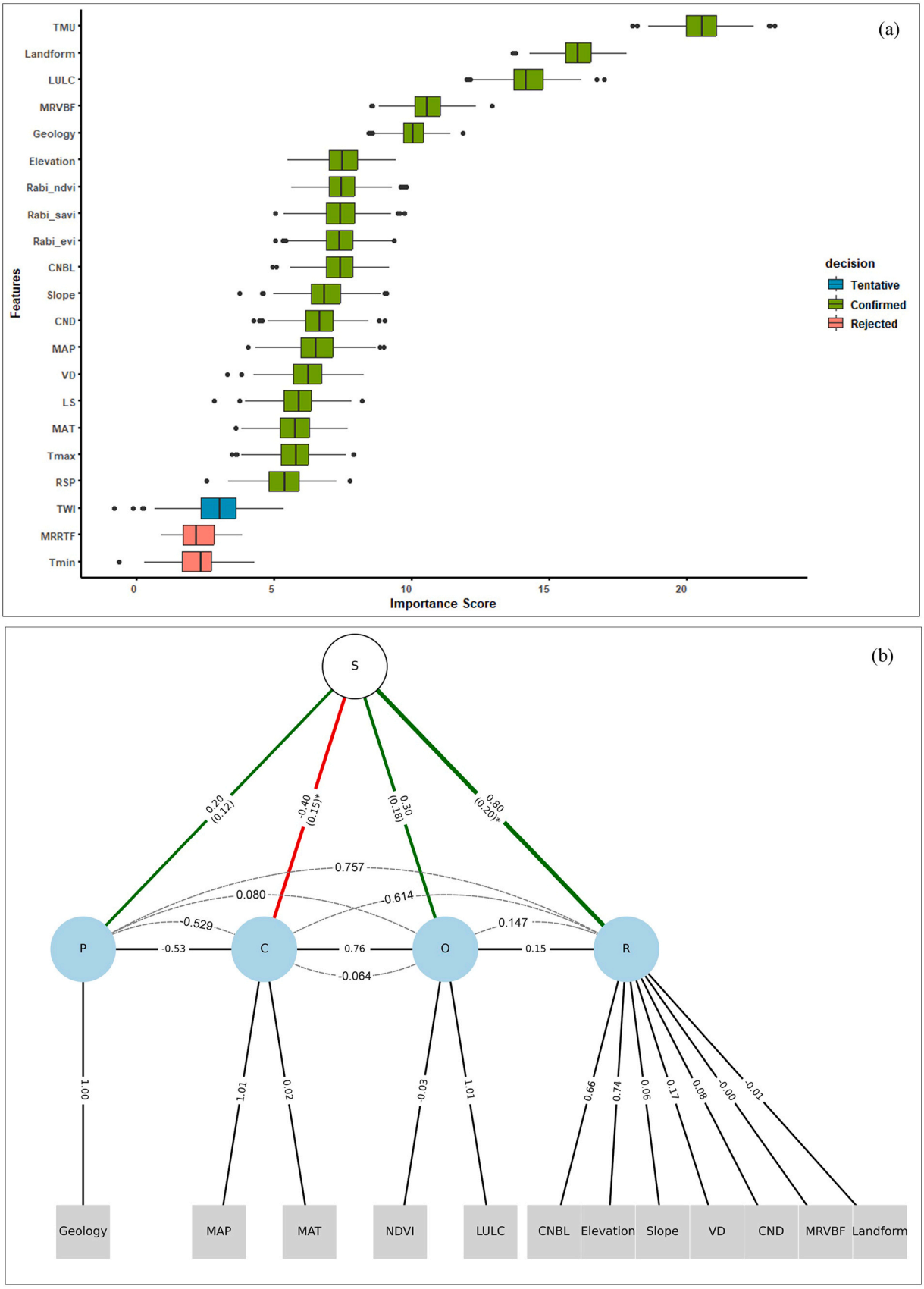


Fig. 6. (a) Selection of important environmental covariates for DSM, based on the *Boruta* algorithm; (b) Bayesian Structural Equation Modelling (SEM) diagram illustrating the direct and indirect effects of Parent material (P), climatic (C), organism (O), and relief (R) as broad factors on soil variability (Sol), with standardized path coefficients representing strength of relationships among selected observed and latent variables considered as sub-factors.

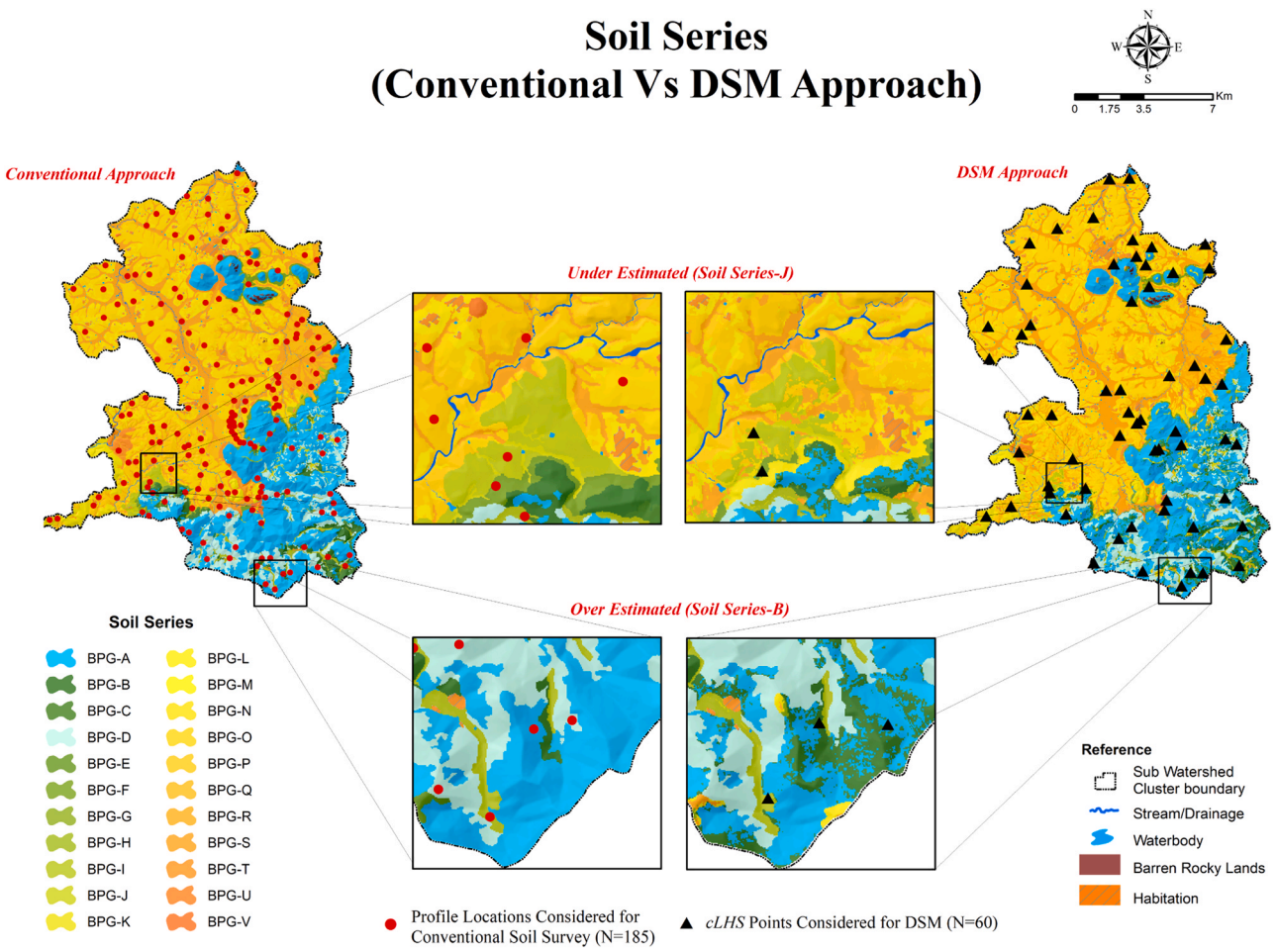


Fig. 7. Comparative visualization of soil series delineation using *Conventional* and DSM approaches, highlighting spatial mismatches.

underrepresented in the DSM output. BPG-C, for example, declined sharply from 110.8 ha to 12.4 ha. BPG-J showed a comparable pattern, with a drop from 174.5 ha to 41.1 ha. These series are often spatially fragmented (very low spatial coverage, even less than 1% of the total geographical area) and occupy transitional or topographically complex landscape positions, where sharp boundaries are difficult to capture with pixel-based classification models. A detailed account of such series-wise overestimation and underestimation is presented in Table 3.

In cases of classification uncertainty, the Random Forest class-membership probabilities (vote proportions) were analysed using Shannon entropy and a confidence (confusion) index (Fig. 8). The entropy map shows that higher uncertainty is concentrated in the northern hilly pockets and parts of the southern forested regions. These areas exhibit higher entropy values (0.30–0.80), reflecting greater ambiguity among competing soil classes. In contrast, the central and relatively well-sampled zones show lower entropy values (nearing 0.10), indicating more stable probability distributions and reduced classification ambiguity. The confidence index map shows a complementary spatial pattern. Higher confidence values (up to 0.98) are predominantly observed in relatively homogeneous and well-represented landscape units, indicating strong dominance of a single predicted class. Conversely, lower confidence values (0.09–0.40) are concentrated in the northern hilly terrain and southern forested tracts, suggesting weaker class dominance and increased confusion among soil series.

Table 3
Comparison of soil series wise area delineation (in ha) between the conventional approach and DSCM-based approach, highlighting over- and under-estimation patterns across Boipariguda Sub-Watershed.

Soil Series	Total Area (ha)		Under-estimation / Over-estimation
	Conventional approach	DSM based approach	
BPG-A	4660	4127	−533
BPG-B	446	833	+387
BPG-C	110	12	−98
BPG-D	1285	1276	−9
BPG-E	14	26	+12
BPG-F	320	266	−54
BPG-G	19	51	+32
BPG-H	7	12	+5
BPG-I	204	196	−7
BPG-J	174	41	−133
BPG-K	127	136	+9
BPG-L	420	928	+508
BPG-M	1881	1874	−6
BPG-N	4758	4595	−162
BPG-O	1296	1013	−283
BPG-P	743	544	−198
BPG-Q	173	51	−121
BPG-R	3059	3764	+705
BPG-S	160	91	−68
BPG-T	163	84	−78
BPG-U	32	7	−24
BPG-V	36	35	−1
Total Area (excluding Miscellaneous)	20092	20092	

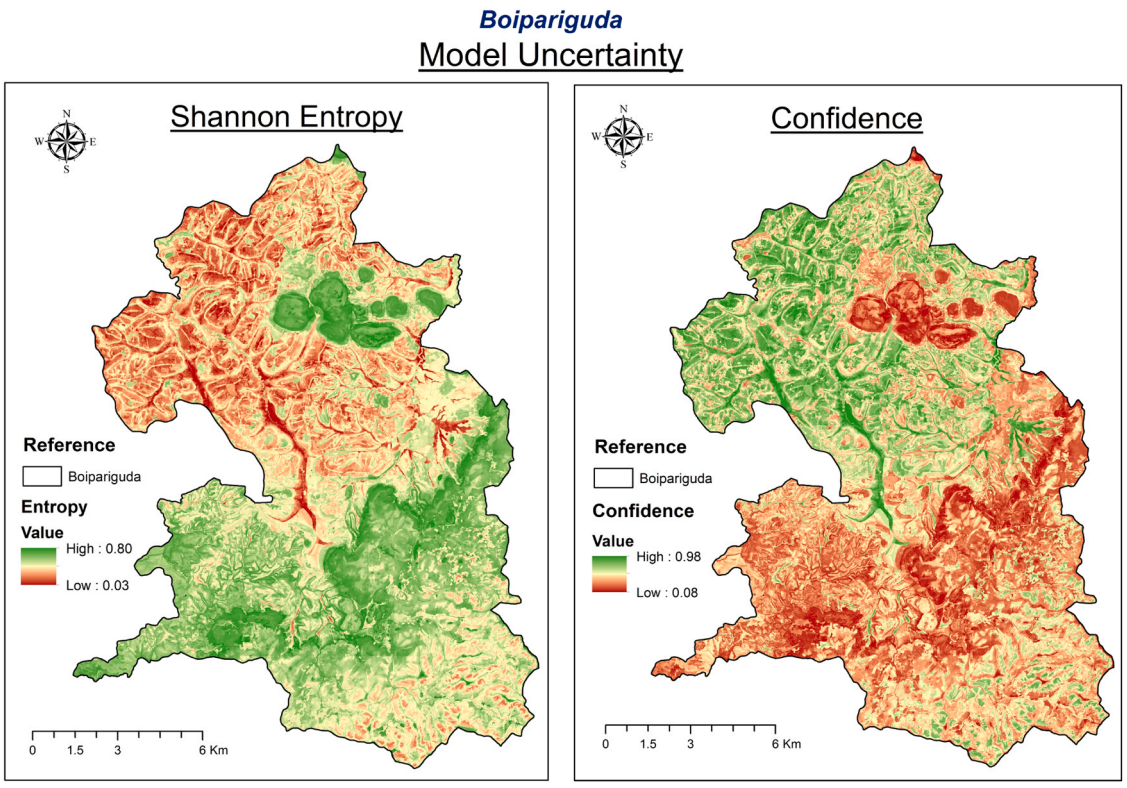


Fig. 8. Spatial distribution of model uncertainty in the Boipariguda region, represented through Shannon entropy and classification confidence.

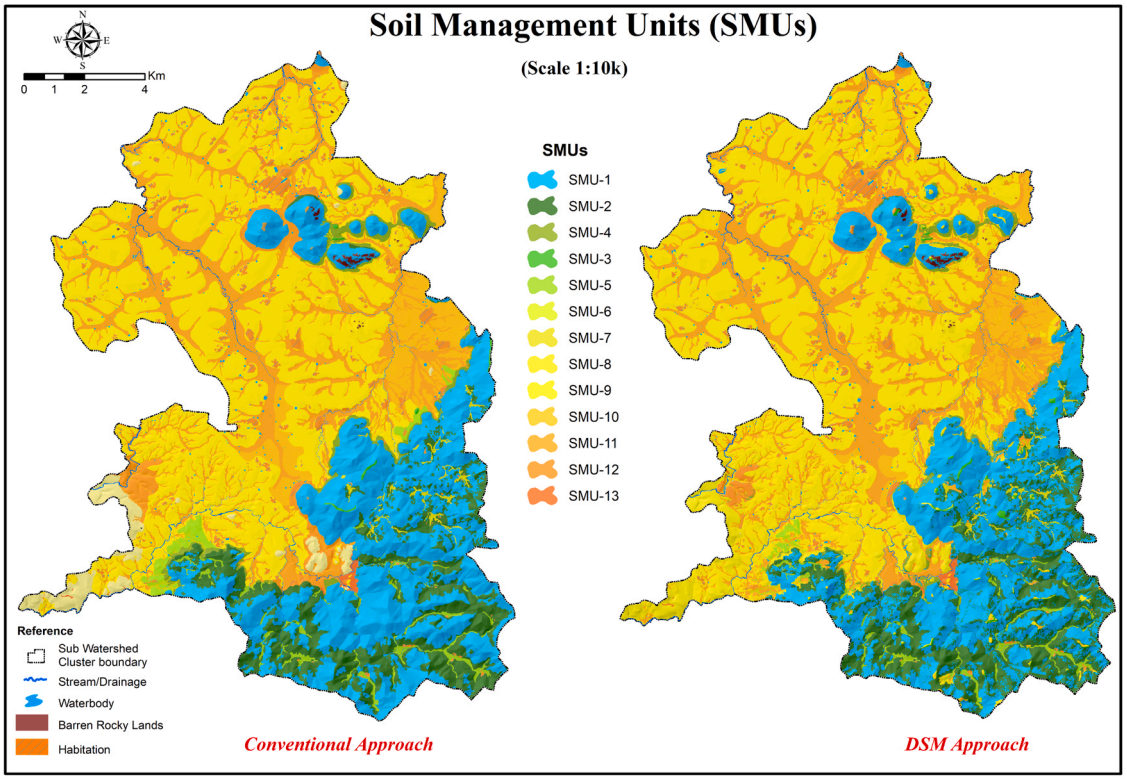


Fig. 9. Comparison of *SMUs* delineated using the conventional *terrain mapping unit*-based *conventional* approach (left) and the DSM approach using RF (right) at 1:10,000 scale.

3.3. From soil series to soil management units: evaluating spatial concordance across approaches

The delineation of *SMUs* from the *conventionally* mapped soil series and those predicted through the RF-based DSM model enabled a practical comparison of spatial correspondence and zone-level integrity (Fig. 9). A total of 13 *SMUs* were derived by grouping pedologically similar soil series with respect to landscape position, morphology, and taxonomic class (Table 4). Soil management units (SMUs) were delineated by merging narrowly differentiated soil series based on the functional influence of soil properties on land use and management response, rather than on minor pedological variations. Grouping was guided primarily by effective soil depth, texture, coarse fragment content, drainage class and soil reaction (pH). The resulting SMUs provide a pragmatic and operational framework for land-use planning in the undulating landscapes of the Eastern Ghats, enabling uniform

Table 4
Distribution of 13 soil management unit over the study area.

SMUs	Soil Series	Area in ha		Over-estimation/ Under-estimation	Description	USDA Soil Taxonomy (WRB)
		Conventional approach	DSM approach			
SMU−1	BPG-A BPG-E	4675	4165	−511	Moderately deep, somewhat excessively drained, moderately gravelly, strongly to very strongly acidic red and lateritic soils with loam to clay loam texture, occurring on hillside slopes with forest, scrub land and waste land under Charnockite landscape.	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
SMU−2	BPG-B BPG-C BPG-D	1843	2114	+271	Deep, well to somewhat excessively drained, slightly to moderately gravelly, moderately acidic, red and lateritic soils with loam to clay loam texture occurring on hillside slope with arable land, plantation and shifting cultivation with intrusion of tree clad area under Charnockite landscape	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
SMU−3	BPG-G BPG-H	26	63	+37	Moderately shallow, somewhat excessively drained, severely gravelly, strongly to very strongly acidic red loamy soils with loam to sandy clay loam texture, occurring on the hill ridge, and isolated hillocks with forest under Charnockite landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
SMU−4	BPG-F	320	266	−53	Very Deep, well-drained, slightly gravelly, strongly acidic, red and lateritic soils with clay loam to sandy clay loam texture occurring on the foothill with arable land with intrusion of forest, waste land, scrub land and tree clad area under Charnockite landscape..	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
SMU−5	BPG-I	204	198	−6	Very Deep, moderately well-drained, non-gravelly, moderately acidic, red and yellow soils with loam to clay loam texture occurring on inter-hill valley of arable land under Charnockite landscape.	Fine-loamy Oxyaquic Haplustalfs (Stagnic Luvisols)
SMU−6	BPG-J	175	41	−133	Very Deep, well-drained, non-gravelly, strongly acidic, red and lateritic soils with loam to clay loam texture occurring on the upland with arable land, plantation, forest, waste land and scrub land under Charnockite landscape.	(Fine Rhodic Paleustalfs) (Rhodic Nitisols)
SMU−7	BPG-K	127	136	+9	Very Deep, somewhat poorly drained, non-gravelly, slightly acidic to neutral, alluvial soils with silty clay loam to silty clay texture occuring on valley with arable land under Charnockite landscape.	Fine Typic Endoaqualfs (Stagnic Gleysols)
SMU−8	BPG-L BPG-U	452	933	+481	Moderately deep to moderately shallow, somewhat excessively drained, moderately to severely gravelly, strongly acidic red loamy soils with sandy loam to loam texture occurring on hillside slopes and sporadic isolated hillocks with forest, tree-clad area, scrub land and waste land under granite-gneissic landscape.	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
SMU−9	BPG-M BPG-N BPG-O	7937	7483	−454	Very Deep, well to moderately well-drained, non-gravelly to slightly gravelly, moderately to strongly acidic, red loamy and red and lateritic soils with clay loam to clay texture occurring on the upland with arable land, plantation and tree clad area with minor intrusion of forest, scrub land and waste land under Granite-gneissic landscape.	Fine Rhodic Paleustalfs (Rhodic Nitisols)
SMU−10	BPG-P BPG-Q	916	597	−318	Very Deep, somewhat poorly to moderately well drained, non-gravelly, slightly to moderately acidic, red loamy and red and yellow soils with silty clay loam to clay loam texture occurring on the plain with arable land with minor intrusion of plantation, forest, scrub land and waste land under Granite-gneissic landscape.	Fine-silty Oxyaquic Haplustalfs (Stagnic Luvisols)
SMU−11	BPG-R	3059	3770	+710	Very Deep, poorly drained, non-gravelly, slightly acidic to neutral, alluvial soils with silty clay to clay texture occuring in the valley with arable land under Granite-gneissic landscape	Fine Typic Endoaqualfs (Stagnic Gleysols)
SMU−12	BPG-S BPG-T	323	177	−146	Very Deep, well to somewhat excessively drained, slightly to moderately gravelly, strongly to very strongly acidic, red and lateritic soils with loam to clay loam texture, occurring on the foothill with forest and tree clad area with intrusion of arable lands patchy under Granite-gneissic landscape	Fine-loamy Typic Rhodustalfs (Luvic Nitisols)
SMU−13	BPG-V	37	36	−1	Very Deep, moderately well-drained, non-gravelly, moderately acidic, red and yellow soils with loam to clay loam texture occurring on inter-hill valley with arable land under Granite-gneissic landscape	Fine-loamy Typic Haplustalfs (Haplic Luvisols)
Miscellaneous		629	629	-		
Grand Total		**20722**	**20722**	-		

recommendations for crop suitability, soil and water conservation, agroforestry and plantation systems. Importantly, the approach retains major soil constraints that directly govern productivity and sustainability at field and micro-watershed scales. Similar terrain-integrated SMU frameworks have proven effective in Indian landscapes for precision land and nutrient management, improving within-unit homogeneity for applied decision-making in semi-arid regions of India (Tiwari et al., 2025). A clear majority-eleven out of thirteen *SMUs*-demonstrated strong spatial agreement between the two approaches. Notable among these was SMU−1, where the mapped area declined slightly from 4675 ha (*conventional*) to 4165 ha (DSM), with minimal loss in hillside gravelly soils. SMU−2, representing deep well-drained arable soils on slopes, expanded from 1843 ha to 2114 ha-an acceptable divergence reflecting improved recognition of transitional terrain. Near-perfect alignment was observed in SMU−5 (204 ha vs. 198 ha) and SMU−13 (37 ha vs. 36 ha), while other well-aligned units included SMU−4, SMU−7, SMU−9, and SMU−11, all showing a minimal deviation. Moderate agreement was observed in SMU−10 and SMU−12, where the DSM outputs underestimated the conventional areas by 319 ha and 146 ha, respectively. These units occupy ecotonal positions such as plains and lower foothills, where variable land use and terrain gradients likely contributed to greater model uncertainty. However, SMU−3, SMU−6, and SMU−8 exhibited substantial spatial deviation and were thus categorized as low-agreement units. SMU−3, representing shallow hill-ridge soils, was overestimated from 26 ha to 63 ha. SMU−6, comprising red and lateritic upland soils, declined from 175 ha to 41 ha, indicating spatial omission in the DSM map. The most pronounced overestimation occurred in SMU−8, which expanded from 452 ha to 933 ha, likely due to confusion between gravelly hillocks and adjacent degraded slopes.

3.4. Accuracy Assessment of DSM approach

3.4.1. Accuracy assessment of DSM-derived Soil Series Map

The RF-based DSM output showed strong agreement with the *conventionally* delineated soil series, as evidenced by both independent point-based validation using 185 soil profiles and pixel-based map comparison across the entire spatial domain. The point-based (independent soil profile-wise) validation of the RF-based DSM soil series map demonstrated an exceptionally high level of accuracy, with most soil series achieving near-perfect agreement (e.g., 17 out of 22 series exhibiting 100% producer's and user's accuracy). This is further supported by an overall accuracy of 97.7% and a Kappa coefficient of 0.904, indicating almost perfect classification consistency with the *conventionally* delineated soil series.

In contrast, the pixel-based (raster) agreement across the entire areal domain (not fully independent owing to shared environmental predictors), is comparatively lower but still exhibits good performance. The overall accuracy of 83.9% and a Kappa of 0.811 suggest substantial agreement. Several soil series such as BPG-K, BPG-V, and BPG-I maintained very high producer's and user's accuracies even at the pixel level, reflecting the model's strong generalization capacity. However, a few series (e.g., BPG-C, BPG-E, BPG-H) showed reduced agreement under pixel-based evaluation, potentially due to limited spatial representation, internal heterogeneity, or overlapping environmental signatures among adjacent soil series. Details are provided in Table 5 and Fig. 10a.

3.4.2. Spatial concordance of DSM- and conventionally derived SMUs

The RF-based DSM approach yielded highly satisfactory classification of SMUs, achieving an overall agreement of 83.9% and a Kappa coefficient of 0.81, indicating strong spatial agreement with *conventionally* delineated *SMUs*. Very high User's Accuracy was achieved for most *SMUs*, including SMU−1 (95.2%), SMU−2 (73.9%), SMU−5 (98.8%), SMU−6 (99.5%), SMU−7 (91.4%), SMU−9 (98.6%), SMU−10 (99.9%), SMU−11 (81.3%), SMU−12 (96.7%), and SMU−13 (99.7%), indicating close correspondence with the conventionally derived SMUs of both valley and upland soils hosting major chunk of cultivated lands. In particular, SMU−5, SMU−6, SMU−9, SMU−10, and SMU−13 demonstrated near-perfect classification, reflecting the model's robustness in well-defined, stable soil-scape settings. SMU−1 and SMU−7 also showed strong, consistent agreement, further supporting the model's ability to capture terrain-driven spatial patterns. A very few *SMUs*, such as, SMU−3 (27.4%), and SMU−8 (46.4%), showed relatively lower User's Accuracy, which may reflect localized landscape heterogeneity, class imbalance, reduced separability among adjacent SMUs, or other model-related uncertainties (Table 6 and Fig. 10b).

Table 5
Accuracy assessment of predicted soil series in sampling point-wise (point-based) and pixel-by-pixel (raster-based) validation approaches.

Series name	Point based			Raster based		
	Producers' Accuracy	Users' Accuracy	Balanced Accuracy	Producers' Accuracy	Users' Accuracy	Balanced Accuracy
BPG-A	87.5	93.3	93.5	85.0	95.2	91.9
BPG-B	83.3	71.4	91.2	59.7	31.7	78.4
BPG-C	50.0	50.0	74.8	5.6	50.1	52.7
BPG-D	100.0	100.0	100.0	100.0	100.0	100.0
BPG-E	100.0	100.0	100.0	15.6	8.2	57.7
BPG-F	100.0	100.0	100.0	64.2	77.1	81.9
BPG-G	100.0	100.0	100.0	79.7	27.4	89.7
BPG-H	100.0	100.0	100.0	42.9	25.3	71.4
BPG-I	100.0	100.0	100.0	96.1	98.8	98.0
BPG-J	100.0	100.0	100.0	23.6	99.5	61.8
BPG-K	100.0	100.0	100.0	98.2	91.4	99.1
BPG-L	100.0	100.0	100.0	99.8	44.4	98.5
BPG-M	95.2	90.9	97.1	87.1	86.4	92.8
BPG-N	84.2	96.7	91.5	89.3	91.6	93.4
BPG-O	90.9	47.6	92.9	46.7	59.2	72.2
BPG-P	93.7	100.0	96.8	72.6	98.6	86.3
BPG-Q	50.0	100.0	75.0	30.3	100.0	65.1
BPG-R	100.0	100.0	100.0	99.4	81.2	97.6
BPG-S	100.0	100.0	100.0	53.8	92.3	76.9
BPG-T	100.0	100.0	100.0	50.6	96.8	75.2
BPG-U	100.0	100.0	100.0	22.8	97.4	61.4
BPG-V	100.0	100.0	100.0	98.6	99.6	99.3
Overall accuracy	97.7%			83.9%		
Kappa	0.904			0.811		

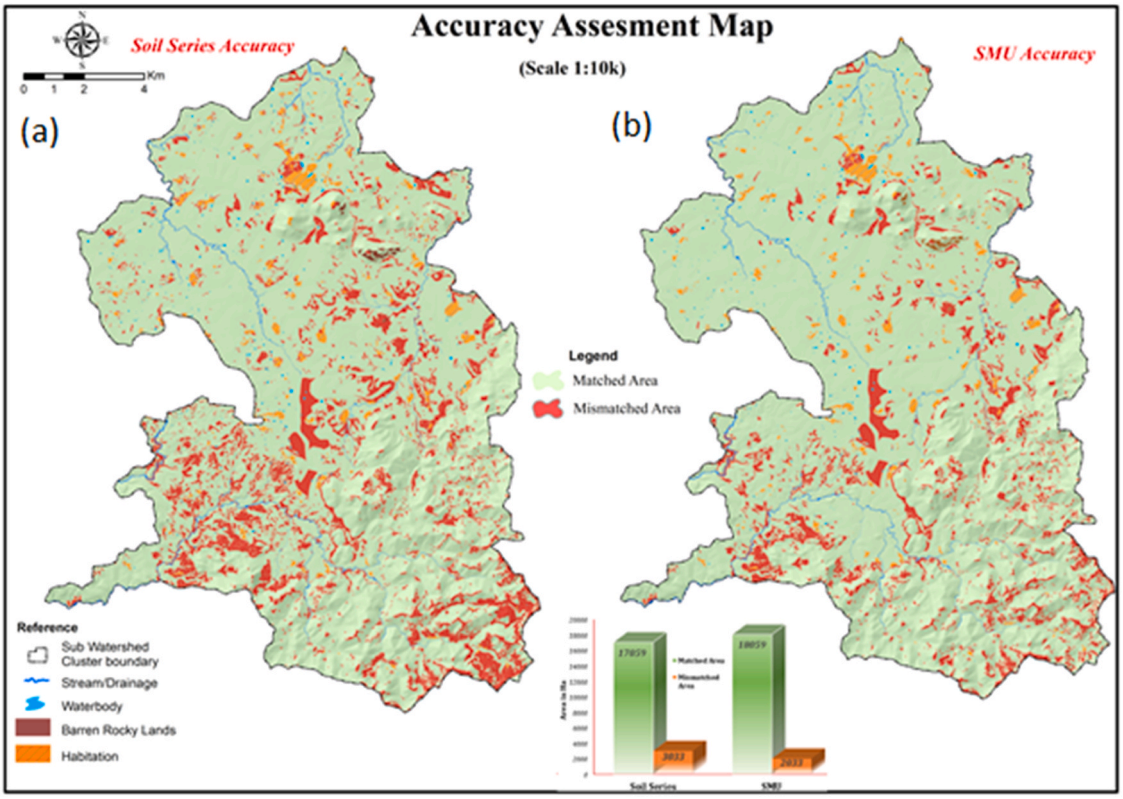


Fig. 10. Accuracy assessment of the RF-based DSM outputs. The top panels display spatial accuracy comparison maps for Soil Series (left) and *SMUs* (right), highlighting matched and mismatched areas between predicted and reference maps. Bar graphs indicate the area under matched and mismatched zones.

Table 6
Accuracy assessment of soil management unit (SMUs).

SMU Class	Producers' Accuracy	Users' Accuracy	Balanced Accuracy
SMU−1	85.2	95.1	97.8
SMU−2	85.6	73.8	91.2
SMU−3	69.8	27.4	84.8
SMU−4	64.2	77.0	81.9
SMU−5	96.1	98.6	98.0
SMU−6	23.6	99.6	61.8
SMU−7	98.2	91.5	99.0
SMU−8	97.6	46.4	97.5
SMU−9	93.8	98.5	96.4
SMU−10	65.5	99.9	82.7
SMU−11	99.4	81.2	97.6
SMU−12	53.4	96.7	76.7
SMU−13	98.6	99.6	99.3
Overall Accuracy			
	89.0		
Kappa			
	0.85		

4. Discussion

This study presents the practical application of a high-resolution (5 m) DSM for detailed soil survey-based soil series and SMU delineation, using the Boipariguda sub-watershed in the Eastern Ghats, characterised by a diverse elevation range, varied land use, and complex geology,-as a representative case study. The work clearly demonstrates that while traditional TMU-based methods remain foundational to soil survey programs, integrating ML models, such as an RF classifier, substantially enhances spatial granularity and classification fidelity.

The initial identification of 22 soil series using *TMUs* reflects an intensive, systematic field-based approach rooted in TMU-soil relationships, capturing the intricate interactions among lithology, topography, slope, and LULC. The clear hierarchical coding of *TMUs* and their integration into field sampling protocols ensures that spatial heterogeneity is embedded from the onset. However, while the *TMU*-based *conventional* framework significantly reduces mapping uncertainty and enhances spatial resolution, it remains constrained by the subjective interpretation of terrain features and manual delineation procedures (Subburayalu and Slater, 2013; Zeraatpisheh et al., 2017). This is particularly critical in unsampled transition zones and heterogeneous units where abrupt or gradual soil changes occur over short distances, limiting the method's sensitivity to sub-unit variation (Rukhovich et al., 2016). In contrast, the *RF classification model-based* DSM approach, trained on environmental covariates and informed by *Boruta*-based variable selection, offered a data-driven process-oriented solution to overcome the spatial limitations of the conventional method, especially in unsampled areas or environments with limited soil information. The significance of DSM lies not only in its ability to replicate the broad spatial trends of soil series but also in its capacity to detect pedogenic gradations within complex terrains (Ma et al., 2019; Adeniyi et al., 2024). The model revealed that several terrain, vegetation, and climate-derived variables played a critical role in soil prediction. *TMU* emerged as the strongest predictor, affirming the relevance of landscape-based stratification, followed by landform, LULC, and MRVBF, which helped capture subtle hillslope curvatures and depositional settings (Chen et al., 2019). Elevation, parent material, and Rabi NDVI also featured prominently, reinforcing the role of topography and seasonal vegetation in influencing soil series variability (Khan et al., 2023; Mehmood et al., 2024).

The Bayesian SEM analysis provides additional insights into soil-*TMU* relationships that remain largely inaccessible through expert-based *conventional* mapping approaches. The dominant direct influence of relief and the complex interactions among elevation, slope, TMU landform, and valley depth quantify relationships that pedologists have long recognised conceptually but have struggled to quantify

systematically (McBratney et al., 2003). The direct positive effects of relief, organism, and parent materials, coupled with negative influences from climate, illustrate the multifaceted nature of pedogenesis, which *conventional* mapping typically addresses through qualitative associations rather than quantitative relationships (Mulder et al., 2017). The Bayesian SEM component particularly distinguishes this approach by providing quantitative insights into factor-level correlations and indirect pathways that influence soil distribution - analytical capabilities that remain largely impossible within *conventional* frameworks. This analytical capability enhances the understanding of soil series occurrence and provides a practical DSM framework for predicting soil type distribution in unmapped areas with similar environmental characteristics, subject to appropriate region-specific model calibration.

The comparative analysis revealed that DSM-derived maps closely matched the conventional series in the majority of cases, particularly for well-defined, large-extent units and arable land. The apparent overestimation and underestimation patterns observed in DSM predictions likely reflect a combination of model uncertainty, class overlap, landscape heterogeneity, and differences in spatial representation between the DSM and conventional mapping approaches, rather than a single underlying cause. The expansion of BPG-B from 446.2 ha to 833.3 ha and BPG-L from 420.1 ha to 928.2 ha likely reflects the model's ability to identify similar pedological conditions across previously unsampled terrain positions, particularly in inaccessible forested areas where *conventional* surveys typically extrapolated limited observations across broad *TMUs* (Mulder et al., 2013). The DSM approach's capacity to distinguish within-forest soil variations represents a particular advancement, as *conventional* soil surveys often characterize entire forested regions as homogeneous units due to sampling accessibility limitations, thereby masking significant pedological diversity. However, the poorer performance on minor soil series exposes an important limitation of the RF model due to class imbalance. Because these series occupied small, fragmented areas and were represented by few calibration profiles, the model had limited capacity to learn their distinctive environmental signatures and consequently favoured better-represented classes. This limitation was further exacerbated in transitional landscape positions, where minor and dominant series shared similar terrain, vegetation, and parent material. Thus, the high overall accuracy was driven mainly by dominant classes and should be interpreted together with class-specific producer's and user's accuracies.

The derivation of *SMUs* from both *conventional* and DSM approaches demonstrates the practical application potential of DSM beyond basic taxonomic classification. *SMUs* represent the operational translation of pedological knowledge into management-relevant spatial units, aggregating similar soil series based on functional characteristics rather than purely taxonomic criteria (Kempen et al., 2009). The strong spatial agreement observed in 11 of 13 SMUs suggests that DSM-derived soil classifications maintain greater spatial integrity for practical land management or crop planning than purely taxonomic classifications at the soil series level.

The integration of soil series prediction with *SMUs* delineation and Bayesian SEM analysis can provide enhanced understanding of soil-*TMU* or landscape relationships. While previous DSM studies have primarily focused on individual soil property prediction or basic taxonomic classification (Minasny and McBratney, 2006), this research demonstrates the feasibility of comprehensive soil type or series mapping at large scale that maintains pedological rigor while enhancing spatial precision and analytical depth. The methodological advances demonstrated in this study have important implications for regional soil mapping and provide a foundation that could support broader national and global soil mapping initiatives through appropriate adaptation and region-specific calibration. The high classification accuracy achieved in croplands (92%) validates DSM's immediate applicability for precision agriculture, while its enhanced discrimination capability in complex terrain positions it as a valuable tool for ecosystem management and large-scale environmental assessment, much awaited to date.

5. Conclusion

This study demonstrates the operational viability of high-resolution DSM for producing detailed soil series and SMUs across undulating and geologically complex landscapes, using the Eastern Ghats of India as a representative case. By benchmarking DSM outputs against conventionally delineated soil series maps, the study provides a systematic assessment of spatial concordance between the two mapping approaches. The Random Forest classifier, supported by Boruta-based variable selection, closely replicated the spatial distribution of major soil series, confirming the model's ability to capture fine-scale soil variability. DSM demonstrated advantages over conventional mapping in areas where traditional surveys are constrained, particularly on steep slopes, in forested terrain, and in transitional zones with sparse sampling. By integrating remote sensing and terrain analytics, DSM filled critical information gaps and represented pedological variability in a more continuous, process-sensitive manner than polygon-based approaches, which often generalise or omit this complexity. In this sense, DSM functions not merely as a digital reproduction of conventional maps but as a diagnostic enhancement, especially valuable for ecologically sensitive and operationally challenging regions. A key methodological advancement is the integration of Bayesian SEM, which revealed the process pathways governing soil distribution.

The delineation of SMUs by grouping functionally and pedologically similar soil series further extends the utility of DSM beyond taxonomic mapping. The strong spatial agreement between DSM-derived and conventionally derived SMUs supports their reliability for large-scale land use planning. The raster-based representation of soil units under DSM offers a more precise and analytically flexible framework for spatial modelling, decision support, and scenario analysis in agriculture, soil conservation, and sustainable land management, particularly under climate change and land degradation pressures. The framework is transferable to regions with similar environmental settings, requiring only minimal recalibration, while areas with contrasting conditions will require context-specific retraining.

Funding Source

World Bank funded Rejuvenating Watershed for Agricultural Resilience (Grant No. NRMANBSSLUPCOl202302500671) through Innovative Development (REWARD), Odisha Project sponsored by DSC&WD, Govt. of Odisha.

CRediT authorship contribution statement

S. Bandyopadhyay: Writing – review & editing, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **S.K. Reza:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **S. Dharumarajan:** Writing – review & editing. **Dalal P. D.:** Visualization, Validation, Software, Formal analysis, Data curation. **S. Paul:** Writing – original draft, Visualization, Software. **B. Das:** Visualization, Validation, Software, Data curation. **S. Dey:** Software. **Rahman F. H.:** Writing – review & editing, Project administration, Funding acquisition. **S. Chattaraj:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors wish to acknowledge Indian Council of Agricultural Research (ICAR) for Institutional Support and Odisha Space Application Centre (ORSAC), Govt. of Odisha for technical support for carrying out the present endeavour as a part of research outcome under the REWARD Odisha Project. We also express our sincere gratitude to the World Bank and the Directorate of Soil Conservation & Watershed Development (DSC&WD), Government of Odisha, for their financial support under Grant No. NRMANBSSLUPCOl202302500671.

Data availability

Data will be made available on request.

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